

InvaSight: an Investment Data Pipeline

Individual client portfolios from four sources, in Saudi Riyal, automatically.

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INVA SIGHT



// what we built

What we built

InvaSight automates the ingestion, standardization and consolidation of investment data from four sources into a single SAR-denominated dataset, replacing slow and error-prone manual portfolio valuation. The finished system runs from source APIs and a synthetic ledger all the way to a live Power BI dashboard, with no manual steps. It has four parts:

- **Automated ingestion.** An internal ledger of synthetic client BUY/SELL trades (50,000 every 45 minutes, about 1.6 million a day), and daily prices for 8 equity tickers (Alpha Vantage), 4 precious metals (metalprice API) and 4 currencies (exchange-rate API).
- **Validation-first loading.** Every batch is checked before it leaves Airflow, lands in Azure Blob Storage, and is loaded into Snowflake one exact file at a time.
- **A SAR-first galaxy schema.** dbt converts every value to SAR and builds three fact tables (fact_holdings, fact_market_prices, fact_daily_portfolio_summary) sharing five dimensions (date, asset, client, currency, source), guarded by 61 tests.
- **A live dashboard.** Power BI connects to Snowflake through DirectQuery: three pages (Portfolio Overview, Allocation & Trading, Market Prices) and 26 measures.

Foundation & architecture at a glance

build & deploy

GitHub Actions builds the Airflow image, tests that every DAG imports, publishes it to Docker Hub and deploys it.

compute

A Kubernetes cluster on AWS EC2 runs Apache Airflow on the master node and dbt on the worker node.

storage & warehouse

Raw files land in the Azure ADLS bronze layer; Snowflake reads and writes Azure data through short-lived SAS tokens.

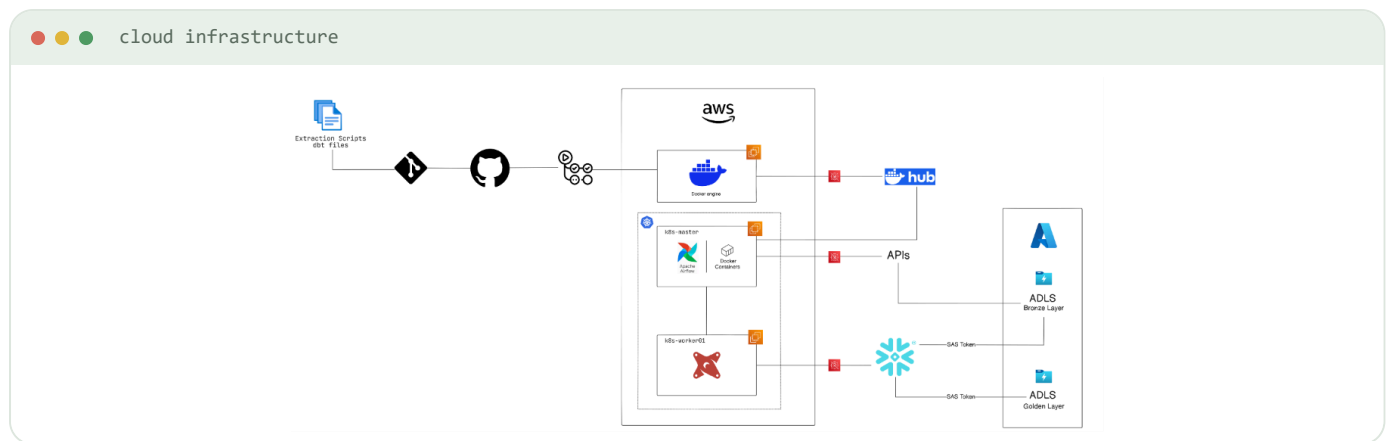


Figure 1. InvaSight cloud infrastructure: GitHub Actions building images, a Kubernetes cluster on AWS EC2 running Airflow and dbt, Azure ADLS storage and Snowflake.

Current status: after the security incident described under *Lessons learned*, the cloud servers were stopped and the pipeline currently runs locally in Docker. The infrastructure will be redeployed with hardened access before it goes back online.

// who it is for

Who it is for

As proposed, the primary users are **wealth management teams managing individual clients' portfolios**, who need a reliable daily view of each client's portfolio in SAR. One daily valuation has to reconcile:

4 sources: a ledger + 3 market APIs	4 currencies: USD · EUR · GBP · CHF → SAR	~1.6M ledger transactions a day	2 calendars: markets close on weekends
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The dashboard gives them each client's portfolio value, P&L, allocation and currency exposure, plus market price trends.

// how it works

How it works

The pipeline follows the proposal's input, processing and output design. Apache Airflow orchestrates every step, and each stage triggers the next in the same run, so there are no schedules to line up.



Ingestion & validation (step 1). `internal_ledger_dag` generates 50,000 transactions every 45 minutes; `daily_multi_resource_dag` fetches equity, metal and FX data daily at 23:00 UTC, using API keys held in AWS Secrets Manager. Each batch is checked (required fields, positive values, expected tickers); an invalid batch stops the run instead of reaching the warehouse.

Landing (step 2). Valid files are uploaded to Azure Blob Storage, in one date-partitioned folder per source.

Loading (step 3). Airflow runs `COPY INTO` to load each new file into Snowflake's RAW layer, one exact file per load, so a retry never loads data twice.

Transformation (step 4). Airflow triggers `dbt build`: staging cleans and deduplicates, every value is converted to SAR, and the analytics layer builds the galaxy schema. All 61 tests must pass for the run to succeed.

Delivery. Power BI reads the analytics tables live through DirectQuery, so the dashboard shows new data as soon as a run finishes.

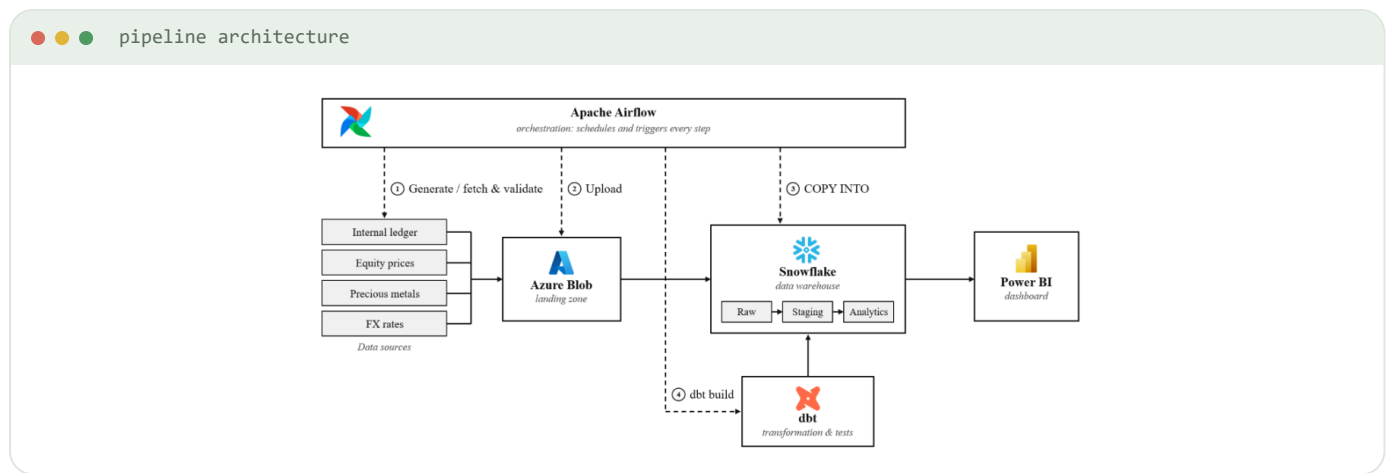


Figure 2. InvaSight pipeline architecture. Apache Airflow orchestrates every step, numbered in run order.

// results

Results

Both DAGs run on schedule and complete end to end (Figure 3). Only validated files land in Azure Blob, a new file for every run (Figure 4). Every build passes all 61 dbt tests (Figure 5), and all 11 SAR-conversion checks pass, with no missing exchange rates and the expected rate for every currency: USD 3.75, EUR about 4.3, CHF about 4.6, GBP about 5.0 (Figure 6). The result is visible in the dashboard (Figure 7).

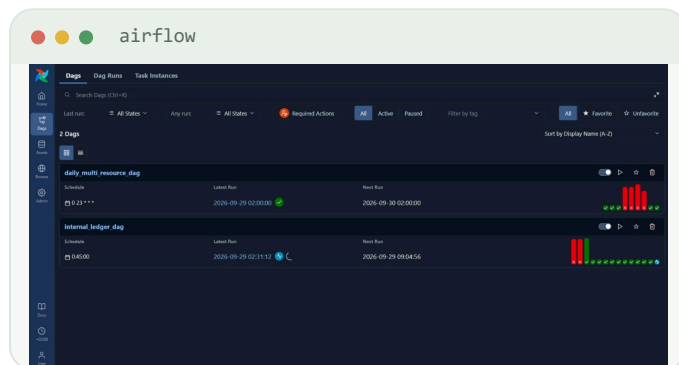


Figure 3. Apache Airflow: the ledger DAG (every 45 minutes) and the market-data DAG (daily) with their recent runs.

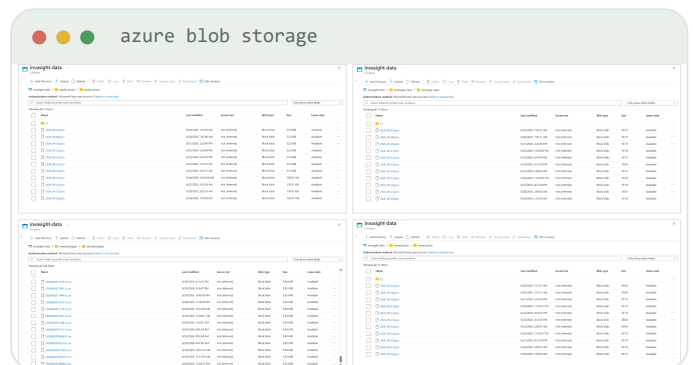


Figure 4. Azure Blob landing zone: equity prices, exchange rates, internal ledger and metal prices, one new file per run.

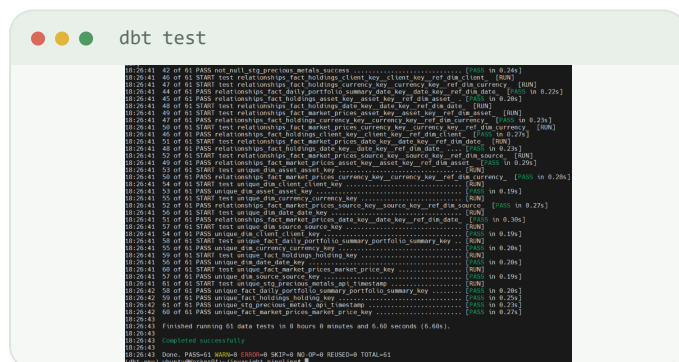


Figure 5. dbt test run: all 61 data tests pass (PASS=61, WARN=0, ERROR=0).

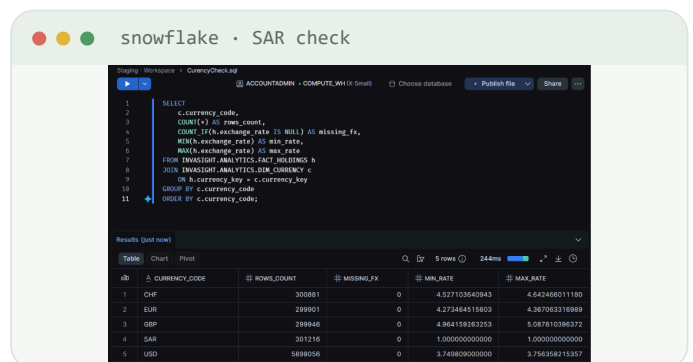


Figure 6. SAR conversion check in Snowflake: no missing exchange rates, each currency at its expected rate.

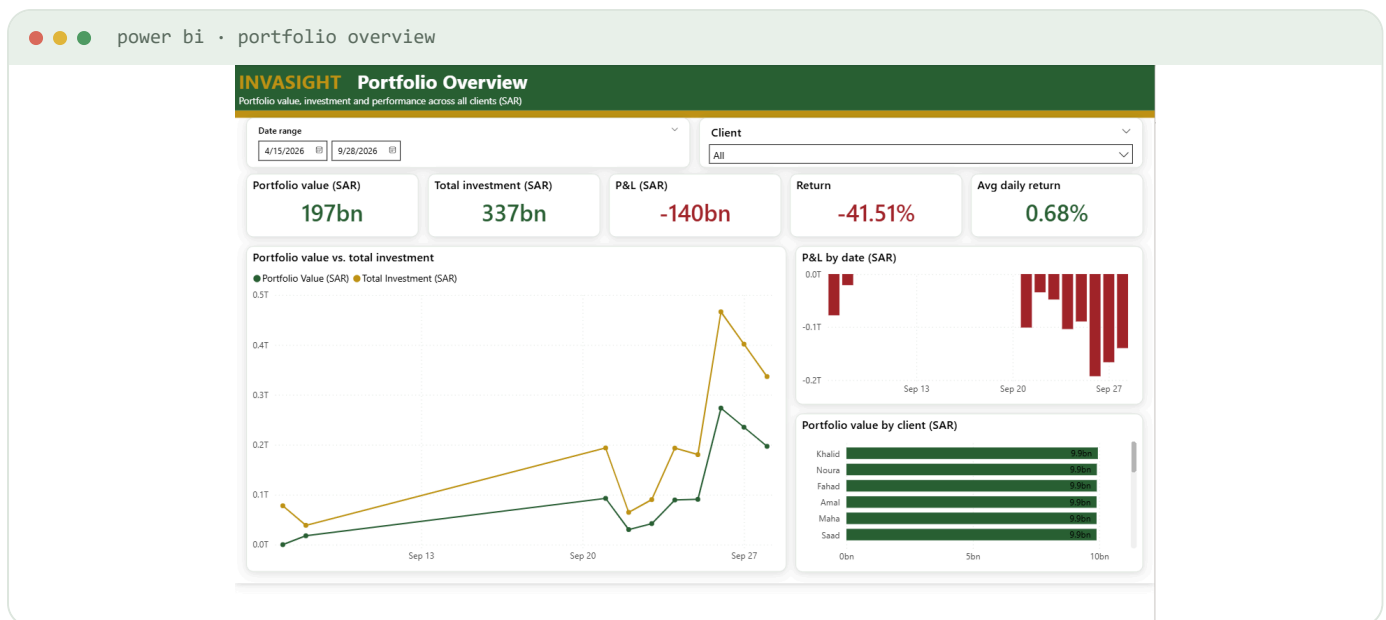


Figure 7. Power BI dashboard, Portfolio Overview page (DirectQuery on Snowflake).

// proposal vs. result

How the result compares with the proposal

proposed	status	notes
Scheduled batch ingestion	✓ done	Two chained Airflow DAGs for all four sources.
Convert all valuations to SAR	✓ done	Single FX-to-SAR model, verified by validation checks.
Align trading calendars	✓ done	Latest rate or price on or before each date.
Galaxy schema	✓ done	3 facts sharing 5 dimensions.
Data quality checks	✓ done	61 dbt tests as a quality gate on every build.
BI dashboard, currency exposure	✓ done	Power BI, 3 pages, 26 measures, FX exposure view.
Asset metadata (sector)	✓ done	Sector classification in dim_asset.
API resilience and availability checks	✓ done	Data validation and API availability checks run before every ingestion.
Azure ADLS for raw data	✓ done	Landing zone for all four sources, one folder per source.
Docker and CI/CD	🔴 partial	The Airflow image is built, tested and published to Docker Hub by GitHub Actions. Separate images for the extraction scripts and dbt, and the Kubernetes deployment through CI/CD, are not built yet.
pytest tests	🔴 partial	DAG-integrity test runs in CI; no unit tests for extraction logic yet.

// worked

What worked well

- **End-to-end automation** of all four proposed sources, from API calls to a live dashboard, with no manual steps.
- **Fresh data in the same run.** Chaining generate → load → dbt build removed a lag of up to 45 minutes for the ledger and about 24 hours for market data.
- **Validated before landing.** A bad batch never reaches the warehouse, and loads are safe to retry.
- **A real quality gate.** 61 automated tests protect the dashboard on every build, and two silent failures (steps reporting success while doing nothing) were found and eliminated.
- **Correct SAR conversion.** All 11 validation checks pass, with no missing exchange rates.

// incomplete

What did not work / is incomplete

- **Synthetic ledger.** Trades are random in direction and price and are not checked against holdings, so portfolio values and returns are illustrative only.
- **Portfolio summary.** Quantities are summed without the BUY/SELL sign and reflect each day's activity. A corrected version (signed quantities, running positions, flow-adjusted returns) was built and validated but not deployed.
- **Market data coverage.** No stock prices on weekends and holidays while the ledger trades daily; new FX loads omit EUR; Alpha Vantage's free tier allows 25 requests a day.
- **Cloud deployment.** The EC2 servers are stopped after the security incident, and CI/CD covers the Airflow image only.

// lessons learned

Lessons learned

- **Check what a green task did.** dbt once reported "Nothing to do" and loads reported success with 0 rows; those steps now fail loudly instead.
- **Chain the stages; don't line up schedules.** Separate schedulers only worked when their timing happened to align.
- **Code that only lives on a server isn't safe.** Git must be the single source of truth.
- **Treat the gold layer as a contract.** Renamed upstream columns broke the DirectQuery dashboard.
- **Secure by default.** An SSH port open to the internet with password login was brute-forced. Hosts now need key-only SSH, restricted security groups, and secrets kept out of code.

// next steps

What we would improve next

- **Complete the CI/CD.** Dockerfiles and images for the extraction scripts and dbt code, deployed to the Kubernetes cluster through GitHub Actions.
- **Harden and redeploy.** Key-only SSH, restricted security groups and rotated credentials, then bring the cloud pipeline back online.
- **A realistic ledger.** Market-based prices, no overselling and lower volume.
- **Correct the portfolio figures.** Deploy the corrected portfolio summary and add a full calendar date dimension.
- **Complete market coverage.** Include EUR in new FX loads and backfill FX history.
- **Engineering hygiene.** Align the dbt project in Git with the deployed models, add unit tests for extraction logic, and adopt incremental dbt models and infrastructure as code.

// in short

InvaSight delivers the core of the proposal: an automated, tested pipeline that consolidates holdings, equity prices, metal prices and FX rates into a SAR-denominated galaxy schema, served live through Power BI. The remaining work is completing the CI/CD, realistic source data and a corrected portfolio summary, so that the dashboard's portfolio figures become as reliable as the pipeline that produces them.

Team & contributions

Fayhaa Alharbi

Airflow orchestration (both DAGs), synthetic ledger generator, API ingestion with validation, Azure landing zone, Snowflake loading and dbt trigger, Docker and CI/CD, Power BI dashboard, recovery of the deployed dbt models, submission assembly.

Maryam Alotaibi

Multi-cloud infrastructure design (Kubernetes on AWS EC2 running Airflow and dbt), security incident response and secrets management, dbt configuration and schema macros, equity and FX staging models, FX-to-SAR conversion, data-quality tests.

Rawan Alaklabi

Analytics layer (galaxy schema of 5 dimensions shared by 3 facts) with tests and documentation, internal-ledger staging model, data-quality tests, ledger sample data, submission folder structure.

Hanoof Alassiri

dbt project setup and environment documentation, precious-metals staging model and tests, Snowflake Azure-stage setup, precious-metals sample data.